A2 · Unit 4 · WBI14 Energy, Environment, Microbiology and Immunity

Decomposition and forensics

Decomposer microorganisms recycle the carbon in dead organisms, and forensic biology uses DNA and the predictable changes in a dead body to link people to evidence and to estimate when death occurred. This page covers decomposition and the carbon cycle, PCR, gel electrophoresis, DNA profiling and the methods for estimating the time of death.

SPEC 6.16 Microorganisms, decomposition and the carbon cycle

WHAT THIS SPEC POINT ASKS

Know how microorganisms decompose organic matter and so recycle carbon.

How decomposers work

- **Decomposers** are bacteria and fungi that feed on dead organisms, faeces and other organic waste (**saprotrophic** nutrition).
- They **secrete enzymes** onto the dead material (**extracellular digestion**). These **hydrolytic** enzymes break large insoluble molecules into small **soluble** ones:
  - carbohydrases (e.g. cellulase, amylase) hydrolyse **glycosidic bonds** → monosaccharides (glucose);
  - proteases → amino acids;
  - lipases → fatty acids and glycerol;
  - nucleases → nucleotides.
- The soluble products are **absorbed** into the microorganisms (by diffusion and active transport).
- They use them in **respiration**, which releases **CO_2** into the air (or into soil and water), and to make their own molecules (proteins, lipids, glycogen) for growth.
- Detritivores (earthworms, woodlice) break dead material into small pieces first, giving a larger **surface area** for the enzymes.

Recycling carbon

- Plants (producers) take in CO_2 in **photosynthesis** and make organic compounds. Animals eat them. When organisms die, the carbon is locked in their bodies.
- Decomposers release that carbon as **CO_2** by respiration, so plants can fix it again. Without decomposers, carbon would stay locked in dead material.
- In **anaerobic** conditions (waterlogged soils, landfill, animal guts), some microorganisms release **methane** (CH_4) instead.

- Decomposition also returns minerals (e.g. nitrogen as ammonium ions from amino acids) to the soil.

What controls the rate of decomposition

Factor	Effect
Temperature	warmer → faster enzyme activity (more enzyme–substrate collisions); too hot denatures enzymes
Water	needed for enzyme reactions and for microbes to grow; very dry → slow
Oxygen	aerobic decomposers are faster; waterlogged soil is anaerobic, so decay is slow
pH	acidic conditions slow most decomposers
Surface area	small pieces decay faster

When decay is very slow (cold, acidic, waterlogged peat bogs), carbon builds up as peat and, over millions of years, fossil fuels (Topic 5).

MARK-SCHEME WORDING

- "Bacteria and fungi are **decomposers**", "**release (hydrolytic) enzymes onto** dead organisms" (2 marks), "break down organic matter", "digested molecules are **soluble**", "**absorbed / taken up** by the microorganisms", "**respired**, releasing **carbon dioxide**", "used by plants for **photosynthesis**".
- 6-mark on carbohydrates: Level 1 = general decomposition; Level 2 = **carbohydrases hydrolyse glycosidic bonds** to monosaccharides, absorbed by diffusion and respired; Level 3 = detail from the data (e.g. one bond broken in a disaccharide, two in a trisaccharide; monosaccharides need no digestion) **and** CO_2 from respiration.
- Faster at 25 °C: "enzymes work faster", "more enzyme–substrate collisions", "mass lost because CO_2 is released by **respiration** of decomposers".
- Products also used to build "**other organic molecules** / glycogen / protein / lipid". Starch or chlorophyll in bacteria isn't accepted.

Often asked as: role of microorganisms in recycling carbon; a 6-mark on decomposing carbohydrates; teabag or leaf mass loss data; plastic breakdown.

COMMON MISTAKE

- A generic decomposition answer when the question is about specific molecules (e.g. carbohydrates): name the enzymes, the bonds, and the products.
- Stopping at "enzymes break it down": say what happens to the soluble products (absorbed, respired, CO_2 released).

- Not linking **mass loss** to CO_2 released by respiration.
- "Food" for decomposers: say sugars or nutrients.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Rate of mass loss from a **tangent** on a decomposition curve (WBI14 Oct 2021).
- Leaves with more sugar decompose faster; bacterial numbers rise then fall as sugars run out (WBI14 Jun 2024).
- Proteins are converted into amino acids and nucleic acids into nucleotides (WBI14 Jan 2026).

SPEC 6.17 PCR: amplifying DNA

WHAT THIS SPEC POINT ASKS

Know how the polymerase chain reaction (PCR) makes many copies of a piece of DNA.

Why PCR is needed

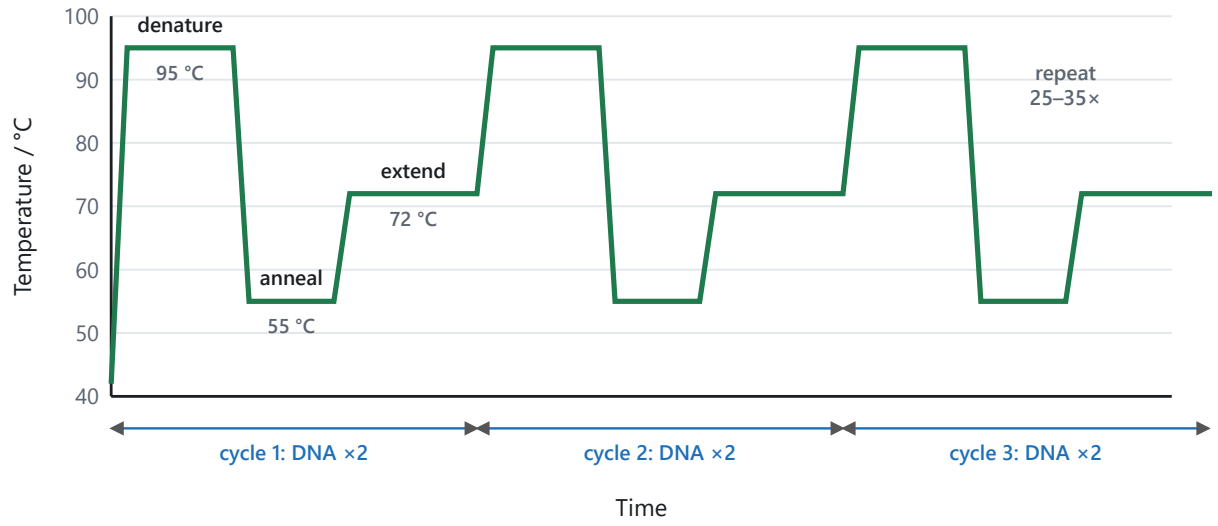
Samples from a crime scene, a fossil, a feather or a hair contain only a **tiny amount of DNA**. PCR copies (amplifies) a chosen section millions of times, so there is **enough to analyse** (e.g. by gel electrophoresis). It happens in a test tube, in a machine called a **thermal cycler**.

What goes in the tube

Ingredient	Job
Template DNA	the sample to be copied
Primers	short, single-stranded pieces of DNA complementary to the start of the target section. **One primer for each strand** (the strands are antiparallel). DNA polymerase can only add nucleotides onto an existing strand, so it needs a primer to start from; primers also choose which section is copied.
DNA nucleotides (A, T, C, G)	the raw material for the new strands
Taq DNA polymerase	joins nucleotides to make new strands (forms **phosphodiester bonds**). From the bacterium *Thermus aquaticus*, which lives in hot springs, so it is **not denatured** at 95 °C.

Ingredient	Job
Buffer	keeps the pH optimal

One cycle



Each cycle runs through three temperatures and doubles the amount of target DNA.

One cycle of PCR

1. **Denaturation (about 95 °C)**: the high temperature breaks the **hydrogen bonds** between the bases, so the two DNA strands separate. (In cells, helicase does this; here heat does.)
2. **Annealing (about 50–60 °C)**: cooling lets the **primers bind** (by complementary base pairing) to the ends of the target section on each single strand.
3. **Extension (about 72 °C)**: Taq polymerase binds at the primer and adds free nucleotides that pair with the template, making a new complementary strand. 72 °C is Taq's optimum.

Repeat **25–35 times**. Each cycle **doubles** the number of copies.

WORKED EXAMPLE

copies and time

Copies after n cycles = starting copies × 2^n.

50 molecules after 25 cycles: $50 \times 2^{25} = 50 \times 33\,554\,432 =$ **1.68×10^{9}** copies.

Time: 30 cycles of (30 s at 95 °C + 30 s at 55 °C + 60 s at 72 °C), plus 2 min at the start and 5 min at the end.

30 × 120 s = 3600 s; 3600 + 120 + 300 = 4020 s = 67 min = **1.12 h** (2 d.p.).

Working out the number of PCR cycles

1. Each cycle doubles the number of DNA molecules, so after n cycles one molecule has become 2^n molecules.
2. To make 128 molecules from one, set $2^n = 128$.
3. $128 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^7$, so 7 cycles are needed.

   Check: after 6 cycles there would be only 64 molecules.

MARK-SCHEME WORDING

- Why: "only **small quantities** of DNA can be collected", so we need to "**amplify** / increase the number of copies of DNA" so there is "enough to run on gel electrophoresis / analyse". "To increase the length of DNA" is not accepted.
- Ingredients: "(DNA) primers", "(DNA) nucleotides", "Taq / DNA polymerase", "buffer". RNA nucleotides or RNA polymerase aren't accepted.
- Cycle, **using the temperatures given**: "94 °C to **break the H bonds** / separate the strands"; "temperature lowered (54 °C) so **primers can attach**"; "temperature raised (72 °C) so nucleotides **join** to the bases"; "many cycles so that enough DNA is made". Maximum 2 marks if you never mention a temperature or temperature change.
- Taq polymerase: "binds to the DNA / primers", "lines up nucleotides along the strands", "forms **phosphodiester bonds**".
- Primers: "so that DNA polymerase can **bind**"; "**one primer for each strand**" because "the strands are **antiparallel** / complementary".

Often asked as: why PCR is needed; list the molecules needed; time for a PCR programme; explain a temperature diagram; why primers are needed; the role of Taq polymerase.

COMMON MISTAKE

- Writing everything you know about PCR and electrophoresis. Use the marks available and the diagram you are given.
- Writing about Taq's heat tolerance when asked for its **role**: it joins nucleotides into a new strand.

- "One primer for each end of each strand." One primer per strand.
- Leaving out the temperatures when a diagram gives them.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

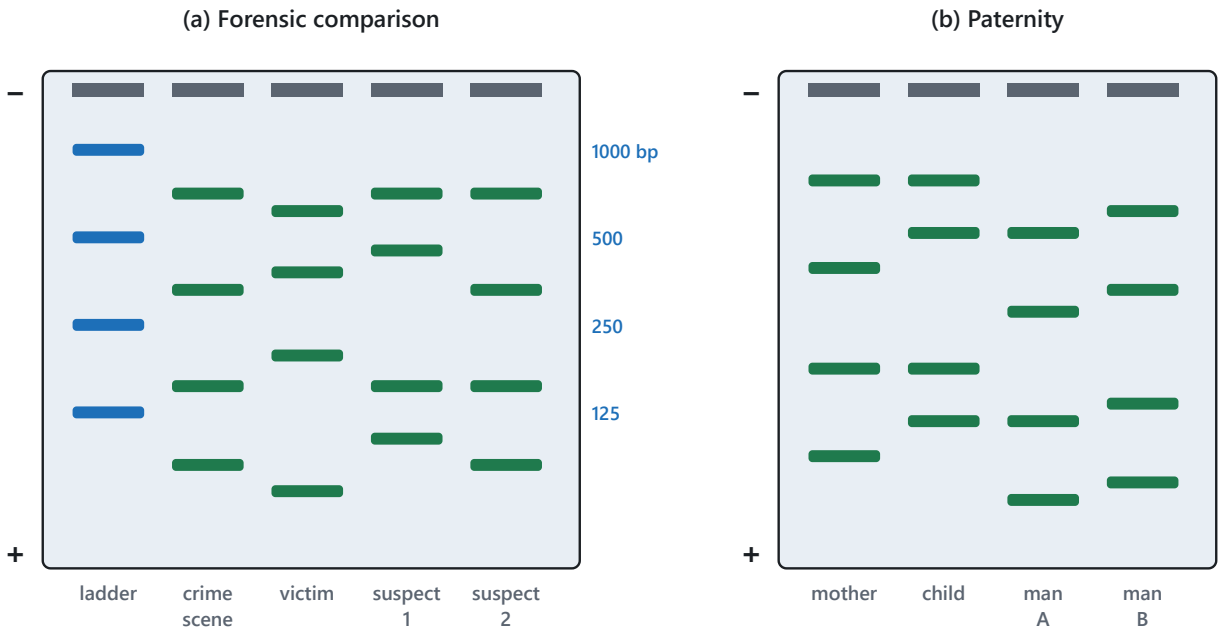
- Reading a PCR temperature programme and calculating its total time (WBI14 Jan 2022).
- Mitochondrial DNA is inherited only from the mother (through the egg), so it traces maternal lines (WBI14 Jan 2026).

SPEC 6.18 Gel electrophoresis

WHAT THIS SPEC POINT ASKS

Know how gel electrophoresis separates DNA fragments according to their length.

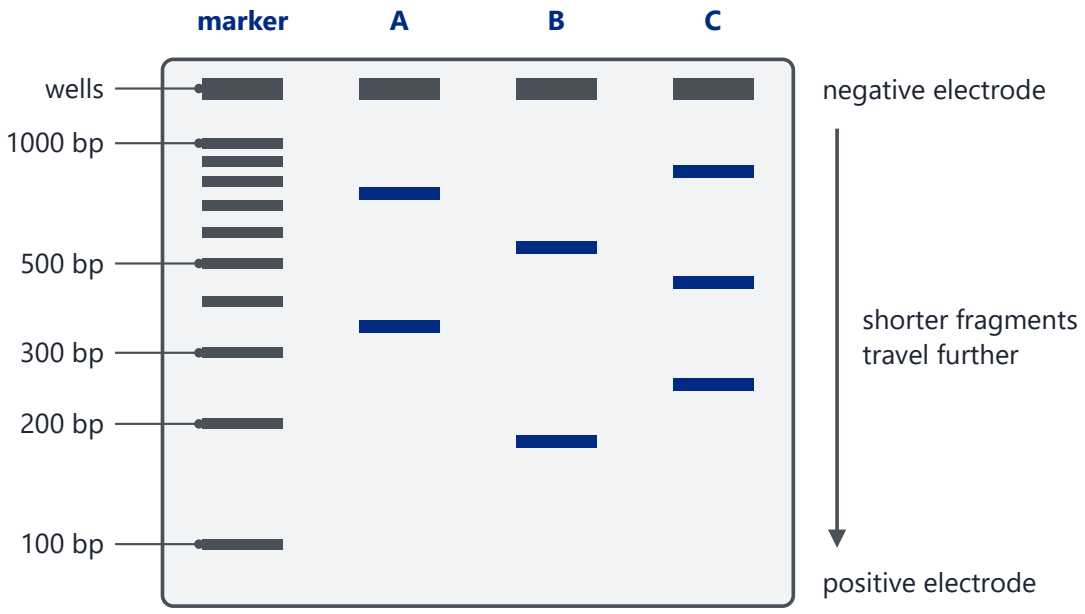
How it works



Wells at the negative end; DNA (negatively charged) moves to the positive electrode. Short fragments travel furthest.

(a) Suspect 2's bands match the crime-scene sample exactly. (b) Every band in the child comes from either the mother or man A.

1. **Get DNA fragments**: PCR products, or DNA cut with **restriction enzymes**. Restriction enzymes cut (hydrolyse the phosphodiester bonds) at specific base sequences called recognition sites. These sites are spaced unevenly, so the fragments are different lengths, and the lengths differ between individuals.
2. **Load**: put the DNA into **wells** cut in a slab of **agarose gel**, covered in a buffer solution. Put a **DNA ladder** (fragments of known lengths) in one well.
3. **Run**: apply a current. The wells are at the **negative** electrode (cathode). DNA is **negatively charged** (its **phosphate** groups), so it moves towards the **positive electrode (anode)**.
4. **Separate**: the gel is a mesh of fibres. **Shorter fragments** slip through the pores more easily, so they travel **further** in the same time. Longer fragments are held back more and stay nearer the wells.
5. **See the bands**: DNA is invisible, so it is stained with a dye that shows under UV light, or labelled with fluorescent or radioactive markers. Each band is many copies of a fragment of one length.



The marker lane gives fragment lengths, so the bands in samples A, B and C can be sized and compared.

- A **higher gel concentration** has smaller pores (more molecules, more resistance), so all fragments move more slowly; it separates small fragments better.

WORKED EXAMPLE

estimating fragment size from a ladder

Ladder bands: 1000 bp at 20 mm, 500 bp at 34 mm, 250 bp at 48 mm from the well. Each halving of size moves the band 14 mm further: distance is linear with **log** of size.

An unknown band is at 41 mm: halfway between 500 bp and 250 bp on the log scale.

log size = log 500 − (7 ÷ 14) × log 2 = 2.699 − 0.151 = 2.548 → size = $10^{2.548}$ ≈ **350 bp**.

(Not 375 bp: halfway on a log scale isn't halfway in base pairs.)

MARK-SCHEME WORDING

- Different fragment sizes: "restriction enzymes recognise **specific (base) sequences**"; "the recognition sites are **not equally spaced**"; "the enzyme **hydrolyses phosphodiester bonds**"; "different-sized fragments **move different distances** through the gel".
- Higher gel concentration: "more molecules / **smaller spaces**, so higher resistance / friction".
- Method details credited: "agarose gel", "apply a current", "use of a dye", "use of restriction enzymes".
- Write "**bands**", not "blocks".

Often asked as: name the enzyme that cuts DNA (MCQ); why fragments differ in size; read sizes off a ladder on a log scale; effect of gel concentration; which lanes are needed to check a GM plant.

COMMON MISTAKE

- DNA moves to the "positive cathode". The positive electrode is the **anode**; the wells sit at the negative cathode.
- Saying larger fragments move further, or explaining by charge alone. All DNA has a similar charge per length; it is the **size** that separates them, because of the gel mesh.
- Writing about introns or hypervariable regions when asked why restriction fragments differ: the answer is unevenly spaced **recognition sites**.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Reading fragment sizes from a ladder plotted on a log scale (WBI14 Jan 2021).
- Gel separation of proteins: thickest band = most concentrated; band that moved furthest = smallest (or most charged) (WBI14 Oct 2022).
- To show a GM broccoli has a gene from linseed, run original broccoli, GM broccoli and linseed side by side; the GM plant must share the linseed band (WBI14 Oct 2024).

SPEC 6.19 DNA profiling: identification and relationships

WHAT THIS SPEC POINT ASKS

Understand how DNA profiling identifies individuals and shows genetic relationships between organisms

(plants and animals).

What a DNA profile is

- Much of the genome is **non-coding**. Some non-coding regions contain short base sequences repeated end to end (**short tandem repeats**, STRs; longer ones are VNTRs).
- The **number of repeats** at each of these places (loci) varies a lot between individuals. So the length of the DNA at each locus varies.
- A DNA profile is the pattern of **bands** made by these fragments on a gel. Apart from identical twins, no two people have the same profile when enough loci are used. You inherit **half your bands from each parent**.

Making a profile: the order of the steps

Making a DNA profile

1. **Collect** a sample containing cells (blood, saliva, hair root, skin, feathers, leaf, droppings).
2. **Extract** the DNA.
3. **Amplify** it with **PCR**: the primers are chosen to flank the repeat regions, so those sections are copied.
4. **Cut** the DNA with **restriction enzymes** (if the method uses them) to give fragments.
5. **Separate** the fragments by **gel electrophoresis**.
6. **Visualise** the bands (dye under UV, or fluorescent/radioactive labels) and **compare** the profiles.

Mark schemes credit "PCR, followed by (gel) electrophoresis" and give credit for restriction enzymes as a detail of the electrophoresis stage. Modern STR profiling often skips the restriction step, because the PCR primers already cut out the right sections. Either way, PCR comes **before** electrophoresis.

Uses

Use	How the profile is used
Forensics	compare crime-scene DNA with suspects and the victim. All bands must match to link a person; one mismatch rules them out.
Paternity / maternity	every band in the child must come from one parent or the other. Bands not from the mother must all be in the father's profile.

Use	How the profile is used
Relationships between animals	the more bands two individuals (or species) share, the more **closely related** they are. Used in conservation **breeding programmes** to choose the **least related** mates, avoiding inbreeding (which makes harmful recessive alleles show up and reduces genetic diversity).
Plants	identify varieties, check that a GM plant carries the new gene, confirm the parentage of crosses, detect illegal timber or fake products
Origins of populations	e.g. show that animals on an island came from a particular mainland population, by comparing profiles

WORKED EXAMPLE

Reading a paternity gel (figure (b) above)

The child has four bands. Two match the mother (top band and third band).

The other two must come from the father. Man A has both; man B has neither.

So man A is the likely father. Man B is excluded.

MARK-SCHEME WORDING

- Relationships: "DNA profiling so that **banding patterns** can be compared; the **more similar** the profiles, the **more closely related**" (or compare base sequences). "Common ancestor" on its own is ignored.
- Method: "take DNA samples from {feather / dropping / skin / blood}", "use of **PCR**", "(followed by) gel **electrophoresis**", "the {number / position / pattern} of **bands** shows similarities".
- Sex and identity from profiles: "bands specific to females on two samples but not on the male's"; "all the animals have **different banding patterns**". "Females will have the same bands" isn't accepted.

Often asked as: show two populations are related; identify individuals and their sex from a profile; compare a museum specimen with a living one; check a GM plant; describe how a profile is made in context.

COMMON MISTAKE

- Describing the method in huge detail and forgetting the context. Name the sample source and say what the bands would show.
- "Blocks" or "lines": write **bands**.
- Saying a child has the same bands as its father. It shares **about half** its bands with each parent.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Mitochondrial DNA (circular, inherited from the mother) compared with nuclear DNA (linear, with histones) (WBI14 Jan 2022).
- Profiling nematodes from a polluted zone and an unaffected area to look for genetic change (WBI14 Oct 2025).
- The chance of two unrelated people matching falls as more loci are used: if 1 in 10 people share a band pattern at each locus, matching at 10 independent loci is 1 in 10^{10}.

SPEC 6.20 Estimating time of death

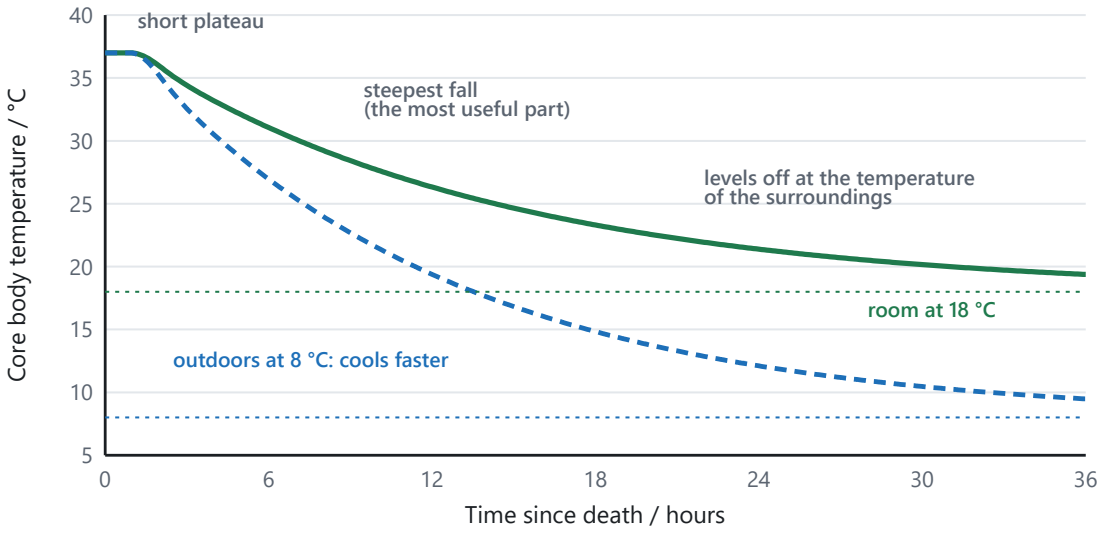
WHAT THIS SPEC POINT ASKS

Understand how time of death of a mammal is estimated from the extent of decomposition, the stage of succession, forensic entomology, body temperature and the degree of muscle contraction (rigor mortis).

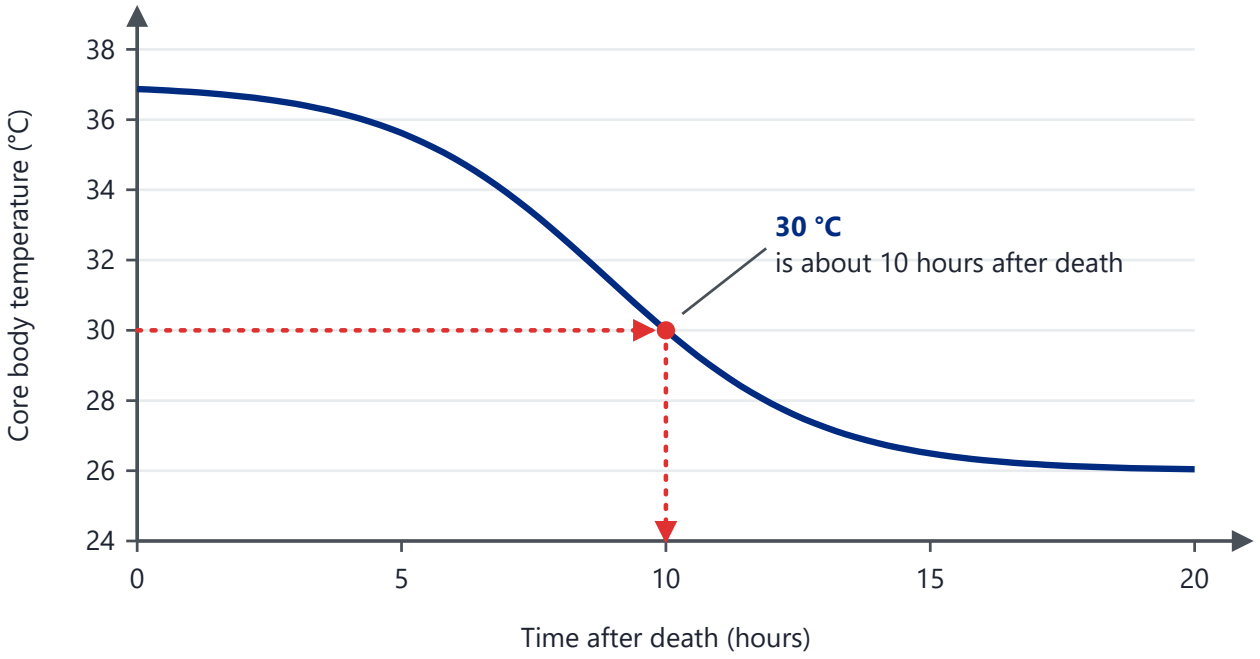
Every method gives an **estimate with a range**, and each is affected by the conditions around the body. Forensic scientists use several methods together.

Body temperature (algor mortis)

- After death, respiration stops, so no more heat is made. The body loses heat to its surroundings until it reaches the surrounding (ambient) temperature.
- Measure **core** temperature with a probe in the **rectum** or **liver** (not the mouth or armpit). Assume it was about **37 °C** at death.
- Cooling is roughly steady (often quoted as about 1–1.5 °C per hour) for the first part, then slows as the body nears ambient temperature.
- Faster cooling (so a *shorter* time since death for the same temperature): cold surroundings, wind, wet clothes or water, a thin body (larger SA : V), no clothes.
- Slower cooling: warm surroundings, clothing, more body fat (an insulator), large body.
- Useful only for about the first **24 hours**: after that the body is at ambient temperature.



Body temperature falls towards that of the surroundings, faster when the surroundings are colder.



A cooling curve: reading across from a measured core temperature gives an estimate of the time since death.

Reading a cooling curve

1. Measure the core temperature, say 30 °C, and the ambient temperature as soon as possible, and choose a curve drawn for similar conditions.

   The rate of cooling depends on the difference between body and ambient temperature, and the ambient temperature may change over time.

2. Find 30 °C on the temperature axis and read across to the curve.

3. Read down to the time axis: about 10 hours after death for the curve shown.

WORKED EXAMPLE

time of death from cooling

One rate. Core temperature 30.4 °C; assume 37.0 °C at death; cooling 1.5 °C per hour.

Fall = 37.0 − 30.4 = 6.6 °C. Time = 6.6 ÷ 1.5 = **4.4 hours** before the body was found.

Two rates (use the table in the question). A body cools 0.8 °C per hour for the first 10 hours, then 0.5 °C per hour. It is found at 25.0 °C.

1. First 10 hours: 10 × 0.8 = 8.0 °C, so 37.0 − 8.0 = 29.0 °C after 10 h.
2. Remaining fall: 29.0 − 25.0 = 4.0 °C; 4.0 ÷ 0.5 = 8 hours.
3. Total = 10 + 8 = **18 hours**.

Muscle contraction (rigor mortis)

- After death, oxygen runs out, so aerobic respiration stops. Cells respire **anaerobically** for a short time: **lactic acid** builds up and the **pH falls**, which inhibits enzymes, so **ATP** can't be made.
- **ATP is needed to detach myosin heads from actin**. Without it the cross-bridges stay locked and the muscles become **stiff**. (Calcium ions leaking from the sarcoplasmic reticulum also let myosin bind.)
- Timing (at room temperature, approximate): starts in small muscles (face, jaw) after about **2–6 hours**; whole body rigid by about **12 hours**; wears off after about **24–36 hours** as enzymes and bacteria break down the muscle proteins.
- Warmth speeds up both onset and loss; cold slows them. Muscle mass and activity just before death also change the timing.

Extent of decomposition

Stage	What happens	Roughly when (temperate conditions)
Fresh	cells' own enzymes start digesting them (**autolysis**)	first day or two
Bloat	gut bacteria decompose tissue **anaerobically**, producing gases (e.g. methane, hydrogen sulfide) that swell the body; skin turns green then purple-black	from about 2–3 days
Active decay	tissues liquefy; gases escape; large **loss of mass**; strong smell	about 1–3 weeks
Advanced decay / dry remains	soft tissue gone; skin, cartilage, then bones	weeks to months; years for a skeleton

- Faster in **warm**, moist, aerobic conditions; slower in cold, dry, buried or underwater bodies.

Succession on a body

- A body is a new habitat. Different organisms arrive in a **predictable order**, and each group changes the conditions, making the body suitable for the next (like succession in 5.15).
- Bacteria first; then **blowflies**, which lay eggs within hours (attracted by the smell); then other flies; then beetles that eat fly larvae and, later, the dry tissue; mites and moths on the dry remains.
- The community present indicates how long the body has been there: useful from days to months and longer.

Forensic entomology

- Identify the **species** of the insects and the **stage** in the life cycle of the oldest ones (egg, larva of a given length or stage, pupa, empty pupal case).
- Each species develops at a known rate **at a given temperature**. Work back from the stage reached to when the eggs were laid. This gives a **minimum** time since death.
- Collect eggs or larvae and **rear them in the lab** at a known temperature to identify the species and check the timing.
- Things that affect it: **temperature** (most important), drugs in the body, humidity, whether flies could reach the body (indoors, buried, wrapped), and that flies lay eggs in daylight. Some species live only in certain habitats, so insects can show that a body was **moved**.
- Most useful after about **3 days**, when temperature and rigor no longer help.

WORKED EXAMPLE

larvae

Blowfly eggs hatch about 24 hours after laying (at this temperature). Larvae are 2.0 mm at hatching and grow 0.20 mm per hour. The longest larva found is 9.6 mm.

Growth time = (9.6 − 2.0) ÷ 0.20 = 38 hours. Add 24 hours as eggs: **62 hours** (minimum time since death).

Which method when?

Time since death	Best methods
up to about 24 h	body temperature; rigor mortis starting
about 3–36 h	rigor mortis (present = between about 3 and 36 h)
days to weeks	forensic entomology; stage of decomposition
weeks to years	succession of organisms; decomposition; skeleton

MARK-SCHEME WORDING

- Measuring: "a thermometer / temperature probe in the **liver / rectum**" (deep in the body). Mouth or armpit is ignored.
- Cooling calculations: a mark for each stage; give the answer to the nearest hour if asked.
- Colder place: the estimate would be **too short** (an underestimate) "because the body **loses heat faster**". Clothing keeps heat in, so the real time is longer.
- Fat: "heat loss would be less / slower as **fat is an insulator**", so the cooling line is shallower.
- Rigor: "**ATP** falls (no aerobic respiration)", "lactic acid rises, pH falls", "myosin heads can't detach from actin"; stiffness alone is limited because "temperature affects rigor", "judging stiffness is subjective", and a stiff body only shows "**between 3 and 36 hours**".
- Body temperature as a method: "readily available", "formulae available", "**but** the change depends on several factors (ambient temperature, clothing, fat, size)", "so limited unless used with other methods".
- Entomology: most accurate after 72 h because "insects colonise a body **in a specific order** / show succession" and their life cycles are known; insects can show the **body was moved** because some species live only in certain **habitats**; collect eggs and rear them "to **identify the species**".
- A cage over a body in a study is "to prevent animals eating / moving it".

Often asked as: where to take the temperature; time-of-death calculations from cooling rates or a cooling graph; why cold surroundings change the estimate; why rigor alone is not enough; forensic entomology and insect succession; evaluate a method (6-mark with chemical markers).

COMMON MISTAKE

- "The body cools faster" when the mark is for "**heat is lost** faster".
- Not using the cooling rates in the table, or using one rate for the whole time when two are given.
- Describing microbial decomposition when the question asks about forensic entomology. Always mention the **species** and the life-cycle **stage**.
- Answering from memory of a similar past question instead of reading the graph you are given.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Fitting a straight line to cooling data (y = mx + c), e.g. a gradient of −1.3 °C per hour (WBI14 Jun 2025).
- Mass of a carcass over time: little change at first; a drop at the end of bloat; a large drop in active decay as gases escape and insects eat the flesh (WBI14 Jan 2022).
- Chemical markers of decomposition (pH, ammonia, hypoxanthine, lactate) in animal models such as pigs; using humans is unethical because the dead can't consent (WBI14 Jun 2023).
- Larval growth rate from data (e.g. 0.12 mm per hour) and the limits of using it (WBI14 Jun 2021).

Quick checks

QUICK CHECK

What is the purpose of heating to 95 °C in each PCR cycle?

- To let the primers bind to the template
- To give the polymerase its optimum temperature
- To break hydrogen bonds so the DNA strands separate
- To join nucleotides to the primers

At 95 °C the hydrogen bonds between complementary bases break, giving single strands that act as templates. Primers bind at the lower annealing temperature, and the polymerase extends them at about 72 °C. The commonest wrong answer confuses denaturation with annealing.

QUICK CHECK

How many PCR cycles are needed to produce 128 molecules from one DNA molecule?

- 6
- 8
- 64
- 7 ✓

After n cycles there are 2^n molecules, and $2^7 = 128$. Six cycles give only 64 and eight give 256.

QUICK CHECK

Why do shorter DNA fragments travel further in gel electrophoresis?

- They pass through the pores of the gel more easily. ✓
- They carry less negative charge.
- They are attracted to the negative electrode.
- They are denser, so they sink further.

Charge per unit length is the same for all fragments, so separation depends on the sieving effect of the gel. DNA is negative and moves towards the positive electrode.

QUICK CHECK

Why do muscles become stiff in rigor mortis?

- Bacteria produce gas inside the muscles.
- ATP is no longer made, so myosin heads stay attached to actin.
- Calcium ions are removed from the muscle.
- The muscles fill with water as cells burst.

After death, aerobic respiration stops; anaerobic respiration produces lactic acid, the pH falls and ATP production stops. ATP is needed to detach myosin from actin, so the cross-bridges stay locked. Gas from bacteria causes bloating, which happens much later.

QUICK CHECK

A clothed body is found indoors. If a cooling curve for a naked body is used, how will the estimated time since death compare with the true time?

- It will be too long.
- It will be correct.
- It will be too short.
- It cannot be estimated at all.

Clothing insulates the body, so it cools more slowly than a naked body. Reaching the measured temperature took longer than the naked curve suggests, so the estimate is too short.

QUICK CHECK

Why must the ambient temperature be recorded when maggots are used to estimate the time of death?

- Maggots only feed on bodies above 37 °C.
- The temperature tells you which species of fly arrived first.
- Cold kills all fly eggs.
- The rate of insect development depends on temperature, because it is controlled by enzymes.

Each stage takes a known time only at a given temperature: warmth speeds development and cold slows it. Without the temperature, the age of the maggots cannot be converted into a time since the eggs were laid.